Lagrange Multipliers, KKT, and Duality

Computer Seminar, Week 6

DSBA

Lecture recap

Use $g_i(x) \leq 0$ and $h_j(x) = 0$:

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \lambda^\top g(x) + \nu^\top h(x), \quad \lambda \geq 0.$$

KKT requires stationarity, primal feasibility, dual feasibility, and $\lambda_i g_i(x) = 0$. The dual function $q(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu)$ gives a lower bound on the primal minimum.

Notebook conventions

Use SymPy for exact algebra, NumPy for numerical checks, and SciPy `minimize` for independent verification. Every answer should report a candidate point, objective value, multipliers, active set, and numerical KKT residuals. SciPy SLSQP uses inequality functions $c(x) \geq 0$. For our convention $g(x) \leq 0$, pass $c(x) = -g(x)$.

Task 1: form the Lagrangian

For

$$\min_{x,y}(x-2)^2+(y-1)^2 \quad \text{subject to} \quad x+y=1, quadx \geq 0, quady \geq 0,$$

write $g_1 = -x$, $g_2 = -y$, $h = x + y - 1$. Build $\mathcal{L}$ in SymPy and differentiate it with respect to x, y. State the sign restrictions on all multipliers.

Task 2: equality multiplier method

Ignore $x, y \geq 0$ temporarily. Solve

$$\partial_x \mathcal{L} = 0, \quad \partial_y \mathcal{L} = 0, \quad h = 0.$$

Check whether the resulting point satisfies the omitted inequalities. If it does, test whether the inequality multipliers can be zero. Explain why the equality multiplier may have either sign.

Task 3: a boundary optimum

Solve

$$\min_{x,y}(x-2)^2+(y-1)^2 \quad \text{subject to} \quad x+y \le 1.5.$$

Consider both cases $g < 0$ and $g = 0$. Reject any candidate that violates feasibility, multiplier sign, or complementarity. Convexity then establishes global optimality.

Task 4: active with zero multiplier

Solve $\min_x x^2$ subject to $x \geq 0$. At the solution, identify the active constraint and its multiplier. The example distinguishes "active" from "positive multiplier." Check

$\lambda g(x) = 0$ numerically.

Task 5: numerical KKT audit

Write a reusable function reporting

$$r_{\mathrm{stat}}, \quad r_{\mathrm{ineq}}, \quad r_{\mathrm{eq}}, \quad r_{\mathrm{dual}}, \quad r_{\mathrm{comp}}.$$

Apply it to exact solutions and SciPy results. Use a stated tolerance to identify active constraints. Explain why a solver's `success` flag and small KKT residuals answer different questions.

Task 6: SciPy verification

Solve the equality and inequality examples with `scipy.optimize.minimize(method='SLSQP')`. Supply analytic objective gradients and constraint Jacobians. Compare the computed point and value with the KKT solution. Reconstruct the multiplier from stationarity, since a SciPy result may not expose it in a consistent form.

Task 7: dual function

For

$$\min_x \tfrac{1}{2}(x-2)^2 \quad \text{subject to} \quad x - b \le 0,$$

derive $q(\lambda)$ by minimizing the Lagrangian over x. Plot $q(\lambda)$ for $b = 1$. Mark the dual maximum and compare it with the primal minimum. Explain why the dual curve never crosses the primal value.

Task 8: multiplier as sensitivity

Solve the scalar problem for several b values below and above 2. Compute a central finite difference for $p'(b)$ and compare it with $-\lambda^*(b)$. Include values near the point where the constraint becomes inactive. State the interpretation in words: a unit increase in the upper bound changes the optimal objective by approximately $-\lambda^*$ units.

Optional extension: two resources

Minimize $\frac{1}{2}\|x - a\|^2$ subject to $x_1 + x_2 \le b_1$ and $x_1 \le b_2$. Vary one bound at a time and compare finite differences with the corresponding multiplier. Notice what happens when the active set changes.

Expected notebook output

- symbolic Lagrangians and solved candidate systems;
- a table of candidate feasibility and active sets;
- KKT residuals for analytic and numerical solutions;
- a primal-dual bound plot and a sensitivity plot;
- short explanations of zero multipliers and the sign of $p'(b)$.

The notebook must run from a fresh kernel after completing the marked cells.

Exit questions

Why does an inactive inequality have a zero multiplier? Can an active inequality also have a zero multiplier? Which assumptions make KKT sufficient? What does a positive multiplier tell us about relaxing a constraint?